

Measurability of the chart integral in the parameter

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Abstract

Astra #10 rank 2. For a coefficient family $c : \Omega \rightarrow \mathbb{N}^2 \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ jointly measurable in (w, s) with a common radius $\rho > b$, the parametrised amplitude $(w, z) \mapsto \Phi_w(z_1, z_2, N z_1^{k_1} z_2^{k_2})$ is measurable: its rectangular partial sums are measurable and converge everywhere on the box, so the limit is measurable (`anaAmp_param_measurable`); the product integrand follows, and the parametric Bochner integral over the box measure gives measurability of $w \mapsto Z_w(N)$ (`twoDamp_param_measurable`), discharging the first measurability hypothesis of the averaged Taylor tree. Zero sorry/axiom.

1 The things formalised

```
theorem anaAmp_param_measurable {Ω : Type*} [MeasurableSpace Ω] (b N : ℝ) (k k' : ℕ)
  (hb : 0 < b) (hb' : b < N) (c : Ω → ℕ × ℕ → ℝ → ℝ)
  (hcm : ∀ ij, Measurable fun p : Ω × ℝ => c p.1 ij p.2) (H : Ω → ℝ → ℝ)
  (hc : ∀ w ij s, |c w ij s| * (ij.1 + ij.2) ^ (ij.1 + ij.2) ≤ H w s) :
  Measurable fun p : Ω × (ℕ × ℕ) =>
    anaAmp (c p.1) b p.2.1 p.2.2 (N * (p.2.1 ^ k * p.2.2 ^ k'))
theorem twoDamp_param_measurable {Ω : Type*} [MeasurableSpace Ω] (b N : ℝ)
  (h h' k k' : ℕ) (hb : 0 < b) (hb' : b < N) (c : Ω → ℕ × ℕ → ℝ → ℝ)
  (hcc : ∀ w ij, Continuous (c w ij))
  (hcm : ∀ ij, Measurable fun p : Ω × ℝ => c p.1 ij p.2)
  (H : Ω → ℝ → ℝ) (hH : ∀ w, Continuous (H w))
  (hc : ∀ w ij s, |c w ij s| * (ij.1 + ij.2) ^ (ij.1 + ij.2) ≤ H w s) :
  Measurable fun w => twoDamp b N h h' k k' (anaAmp (c w) b)
theorem twoDamp_param_aestronglyMeasurable ... :
  AEStronglyMeasurable (fun w => twoDamp b N h h' k k' (anaAmp (c w) b))
```

2 Mathematics

Each term $c_w(ij, s) \bar{u}^i \bar{v}^j$ ($\bar{u}, \bar{v}$ the clamped coordinates, $s = Nu^{k_1} v^{k_2}$) is measurable in (w, z) ; the finite partial sums over $F \subset \mathbb{N}^2$ are measurable; for every (w, z) the series is absolutely convergent since $\bar{u}, \bar{v} \leq b < \rho$ and $|c| \rho^{i+j} \leq H$, so the partial sums converge to the amplitude along the directed family of finite sets, a countably generated filter; hence the amplitude is measurable as a pointwise limit. The integrand $u^{h_1} v^{h_2} e^{-\beta s^2} \Phi_w$ is then measurable on $\Omega \times \mathbb{R}^2$, and the integral over the finite box measure is measurable in w by the Fubini-type measurability of parametric integrals; finally $Z_w(N)$ equals that integral by the product-form identity of unit 89.

3 Paper-facing meaning

Ordinary but necessary: the per-stratum integral $\int Z_w(N)\rho_I(w)dw$ of `thm:TaylorTree` is well defined once the chart amplitude depends measurably (in the natural joint sense) on the noncritical coordinates, which holds for analytic ξ, η . Together with unit 137 this makes the averaged Taylor tree a theorem about a measurable family with a common envelope, modulo the measurability of the canonical coefficients (Astra #10 rank 3, next).

4 Lean notes

- `measurable_of_tendsto_metrizable'` `atTop` works along the filter of finite subsets of $\mathbb{N}^2$ because `atTop.isCountablyGenerated` applies to any countable preorder; the `HasSum` of each point is exactly the required `Tendsto` of partial sums, packaged with `tendsto_pi_nhds`.
- `StronglyMeasurable.integral_prod_right'` needs `SFinite` of the second measure; for the project's `boxMeasure` (a plain `def`) provide it with a local `have ... := by unfold boxMeasure ; infer_instance`.
- The linter prefers `have` to `haveI` for propositional instances.